2. Central Tendency Distortion (Shape Stability)

This component measures whether the return distribution remains **centered and structurally stable**, or whether it exhibits **internal drift away from its robust center**.

It captures **shape deformation inside a rolling window**, independent of directional imbalance (handled by symmetry).

📌 Purpose

To quantify whether price behavior is:

- **centered and stable → equilibrium-like**
- **shifted or structurally biased → regime formation / deformation**

It detects **internal distribution drift**, not direction.

🧮 Definition

$$C_t = \frac{|\mu_t - m_t|}{S_t}$$

Where:

- μ_t = mean of returns in the rolling window
- m_t = median of returns in the rolling window
- S_t = rolling movement scale (defined below)

📏 Scale (Movement Energy Baseline)

$$S_t = \mathrm{E}_{t-k:t}\left[|close_t - close_{t-1}|\right]$$

Implementation:

```
df["step_move"] = df["close"].diff().abs()
df["scale"] = df["step_move"].rolling(scale_window).mean()
```

🔑 Meaning of Scale

This is not volatility in a statistical sense.

It represents:

> The **average realized price movement per step under current market conditions**

It is a **local movement energy baseline**, which adapts to regime changes.

📊 Interpretation

C Value	Meaning
~0	distribution is centered and stable
low	minor structural distortion, equilibrium-like
moderate	emerging internal drift or structural bias
high	strong deformation of distribution center relative to movement energy

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🧠 What This Measures

This metric answers:

> “Is the center of the return distribution stable, or is it drifting relative to how much the market typically moves?”

It captures:

- internal skew formation
- early structural drift before directional imbalance appears
- hidden regime transitions
- asymmetry in distribution shape independent of order-flow symmetry

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⚠️ Key Distinction

This is NOT symmetry.

Component	Meaning
Symmetry (S)	directional imbalance (buyers vs sellers)
C (this metric)	internal structural distortion of distribution center

A market can be:

- symmetric but structurally drifting
- asymmetric but centrally stable
- or both unstable depending on regime phase

🧩 Role in System

Central Tendency Distortion acts as a **structural early-warning signal**:

It helps identify:

- pre-trend deformation (before directional imbalance emerges)
- distribution drift under still-balanced order flow
- regime transitions masked as equilibrium
- divergence between statistical center and robust center

🧠 Key Insight

This metric is best interpreted as:

> **"Structural drift per unit of current market movement energy."**

It separates:

- real regime formation
  vs
- noise-induced statistical fluctuations